Preliminaries: Continuity and Limits - Intuition

Previously, we said that a function is continuous at a point $x = c$ if both the following conditions are satisfied:

- We know what number we expect $f(x)$ to take at $x = c$; and
- $f(x)$ is actually that number when $x = c$.

Three Ways This Can Go Wrong

Limits – A Better Definition

Definition $\lim_{x\to c} f(x) = L$ if we can get $f(x)$ as close to L as we want by taking x close to c. (In words, we say "the limit as x goes to c of $f(x)$ equals L.")

- In this situation, as we get very close to c from the horizontal direction, the graph of $f(x)$ gets very close to L in the vertical direction.
- Also notice that $x \to c$ means that x gets very close to c, but is not equal to c.
- We say $\lim_{x\to c} f(x)$ *does not exist* if there is no such number L.

Definition (Left and Right Limits)
$\lim_{x\to c^+} f(x)$ means that we only let x approach c from the *positive side*. In terms of the graph, x is approaching c from the *right*.

$\lim_{x\to c^-} f(x)$ means that we only let x approach c from the *negative side*. In terms of the graph, x is approaching c from the *left*.

Important point: $\lim_{x\to c} f(x)$ exists and is equal to L if and only if

$$\lim_{x\to c^+} f(x) = L = \lim_{x\to c^-} f(x)$$

1. By looking at $\lim_{x\to 0^-} f(x)$ and $\lim_{x\to 0^-} f(x)$, investigate $\lim_{x\to 0} f(x)$ where

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(a) $f(x) = \frac{|x|}{x}$.

(b) $f(x) = \frac{1}{x}$

(c) $f(x) = \frac{1}{x^2}$

Limit Laws

Here are properties that allow us to easily calculate limits. Note: To use them, $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ must exist.

- $\lim_{x\to c} k =$ _____ for any constant k.
- $\lim_{x\to c} x =$ _____.
- $\lim_{x\to c}[kf(x)] =$ _____ for any constant k.
- $\lim_{x\to c}[f(x)+g(x)] =$ _____.
- $\lim_{x\to c}[f(x)g(x)] =$ _____.
- $\lim_{x\to c}\left[\dfrac{f(x)}{g(x)}\right] =$ _____ if $\lim_{x\to c} g(x) \neq 0$.

Another Limit

Definition Limits at Infinity

We say $\lim_{x\to\infty} f(x) = L$ if we can get $f(x)$ as close to L as we please by taking x _____.

Using Limit Laws to Calculate Limits

Let's compute $\lim_{x\to\infty} \dfrac{3x^2+17x-7}{2x^2-1}$

We already know this is $\frac{3}{2}$ from our time with rational functions, but let's do it from scratch using limit rules:

More Limits

In Questions 2-5, you will be asked to calculate limits. You may use the limit laws above, but you should also try plugging in first. If you get a number, that is your answer. If you get $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then you will need to do some algebra to simplify the expression before calculating the limit. You may not use any techniques we have not yet covered in this class so far.

2. $\lim_{x \to 5} \frac{x^3 - x}{2x + 3}$

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3. $\lim_{x \to 9} \frac{\sqrt{x} - 3}{x - 9}$. (Reminder: no techniques we have not yet covered in this class so far!)

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4. $\lim_{h \to 0} \frac{2(h+2)^2 - 2h^2}{h}$ (Hint: simplify first. You may need to refer to the limit rules above and to question 1.)

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5. Let

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$$f(x) = \begin{cases} x - 5 & \text{if } x \geq 1 \\ -4x & \text{if } x < 1. \end{cases}.$$

Evaluate $\lim_{x \to 1} f(x)$.

Continuity and Limits

Definition (Continuity) A function is said to be **continuous** on an interval $[a, b]$ if, intuitively, you can draw the graph of f over that interval without lifting your pencil from the paper.

The mathematical definition is that a function f is continuous at the point $x = c$ if

$$\lim_{x \to c^+} f(x) = \lim_{x \to c^+} f(x) = f(c).$$

The first part of the equality says that the limit exists (that is, we know what value to expect at $x = c$). The second equality says that we also *get what we expect.*

6. Consider the following function:

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$$f(x) = \begin{cases} kx & \text{if } x \geq 1 \\ x^2 + k^2 - 1 & \text{if } x < 1. \end{cases}.$$

(a) Find $\lim_{x \to 1^+} f(x)$ and $\lim_{x \to 1^-} f(x)$.

(b) What is $f(1)$?

(c) What must be true about your three answers in order for $f(x)$ to be continuous? Find all values of k that make it so.